EPFL — Fall 2026

COM-309

Introduction to quantum information processing

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This document was originally created for my own use, but I am sharing it because I believe it may be useful to others for studying and reference purposes. However mistake can happen and there will inevitably be some errors, if you found any, you are more than welcome to send me a message (Arthur Hbe on telegram) or to directly do a PR on the repo

`https://github.com/Arthur926564/LectureNotes`

If you think that something is missing or that something is not clear, I will be more than happy to add your modification to this document.

Unless otherwise stated, all material in these notes is based on the course content, including the diagrams, explanations, and examples.

Note also that these notes were taken during the lecture, but not only; some materials was added later (that's why you may read remarks that sounds like I knew the course before taking it).

Finally, thanks a lot to Joachim Favre for his help with the setup and whose style is used in this document.

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Introduction

Overview of the course

Quantum information processing is a sort of unification of the two followings fields: **physics** and **computer science**/ **communication**. Quantum physics is a new field of physics that emerged around 1900/1930, we found that that the behaviour of particle at a really small scale was really different of what we expected, uncertainty was the new discovery at atomic scale. We are not able to exactly predict how a particle behave even by knowing every information that could possibly be known.

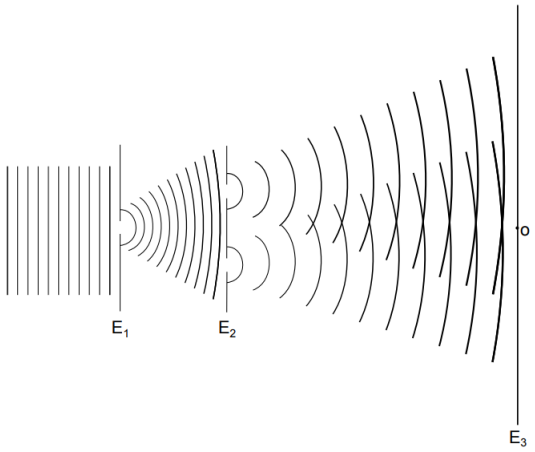
Documents

The class materials are typed into lecture notes which covers this course and the quantum computation course CS-308 (chapters 9, 10, 11, 12, 13, 14).

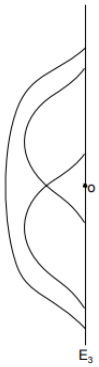
Chapter 1

Wave-Particle Duality

Interference in Double slit exp.



What this has shown is that light behaved like a wave in our case. If light were in fact particle, the result of the following experiences would be different. The resulting wave would be something like this:



However lights behaving like a wave is something that we can all agree on, on the other case, imagine doing the same experiment but for electrons or C60, the resulting frequency would corresponds to a wave as for lights. And this frequency is **constant** i.e., we can send electrons one after the other then after collecting the statistics; remarkably the function we get is the same prediction as the one of wave.

Chapter 2

Mathematical notions

2.0.1 Recap of linear algebra in Dirac notation

The arena for quantum mechanics is Hilbert space (a vector space on $\mathbb{C}$).

Notation

The language in order to speak in quantum science is **Dirac** notation.

Definition 1. Ket

$$\vec{\varphi} = |\varphi\rangle = \begin{pmatrix} \varphi_1 \\ \varphi_1 \\ \vdots \\ \varphi_n \end{pmatrix} \quad , \varphi_i \in \mathbb{C}$$

Definition 2. Bra

$$\vec{\varphi}^\dagger = \langle\varphi| = (\varphi_i^*, \varphi_2^*, \ldots, \varphi_D^*), \varphi_i \in \mathbb{C}$$

Kets and Bras are the foundation of the language of quantum science, those will be used extensively throughout the whole course.

Definition 3. Inner product In our new space, we will define the inner product as

$$\vec{\varphi}^\dagger \cdot \vec{\varphi} = \sum_{i=1}^{D} \varphi_i^* \cdot \varphi_i$$

Properties

As for vector in classical vector space, vector inside Hilbert space also end up having properties that we can use in order to prove theorems:

$$\langle\varphi||\varphi\rangle^* = \langle\varphi||\varphi\rangle$$
$$(\alpha^*\langle\varphi_1| + \beta^*\langle\varphi_2|)|\varphi\rangle = \alpha^*\langle\varphi_1||\varphi\rangle + \beta^*\langle\varphi_2||\varphi\rangle$$

The positivity of bracket:

$$\langle\varphi||\varphi\rangle \geq 0$$

We also have triangle inequality and Cauchy-Schwartz:

$$\|\varphi + \varphi\| \leq \|\varphi\| + \|\varphi\|$$
$$|\langle\varphi||\varphi\rangle| \leq \|\varphi\|\|\varphi\|$$

Orthonormal basis in $\mathcal{H}$

the dimension of our space is denoted as $\dim\mathcal{H} = D$. The basis of that space is composed of basis vector which are noted as:

$$|u_1\rangle, |u_2\rangle, \ldots, |u_D\rangle$$

Where

$$\langle u_i||u_j\rangle = \begin{cases} 1 & i = j \\ 0 & j \neq i \end{cases}$$

as vector that are not equal are orthogonal hence their inner product equals 0.

Example

The most commons space that will be used during this course is $\mathbb{C}^2 = \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \alpha, \beta \in \mathbb{C} \right\}$. In this case we define our vector as

$$|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Tensor product

The goal of a tensor product is to combine spaces together.

Definition 4. Given $\mathcal{H}_1$ and $\mathcal{H}_2$ we define $\mathcal{H}_1 \otimes \mathcal{H}_2$ as Where $i = 1, \ldots, D_1$ and $j = 1, \ldots, D_2$.

Example

Taking again $\mathbb{C}^2$ We want to convert it into $\mathbb{C}^2 \otimes \mathbb{C}^2$,the new basis of our space this gives:

$$\begin{aligned} |0\rangle \otimes |0\rangle &= |0\ 0\rangle \\ |0\rangle \otimes |1\rangle &= |0\ 1\rangle \\ |1\rangle \otimes |0\rangle &= |1\ 0\rangle \\ |1\rangle \otimes |1\rangle &= |1\ 1\rangle \end{aligned}$$

This allows us to have multiple qbits inside one state, which later on will allow us to have the notion of entanglement.

If we wanted to translate this into classical linear algebra, this result for the tensor product would be the following: given two vectors $\vec{v} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $\vec{u} = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ the tensor product would be computed as

$$\vec{v} \otimes \vec{u} = \begin{pmatrix} \alpha \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \\ \beta \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}$$

Inner product in $\mathcal{H}_1 \otimes \mathcal{H}_2$

$$(\langle\varphi_1| \otimes \langle\varphi_2|)(|\varphi_1\rangle \otimes |\varphi_2\rangle) = \langle\varphi_1||\varphi_1\rangle\langle\varphi_2||\varphi_2\rangle$$

Matrix

Definition 5. A matrix $A : \mathcal{H} \to \mathcal{H}$ is an operator defined in a given an orthonormal basis $\{|1\rangle, |2\rangle, \ldots, |D\rangle\}$ which can be represented as an $D \times D$ array where $A_{ij} = \langle i|A|j\rangle$. We can even define our matrix as:

$$A = \sum_{i,j=1}^{D} A_{ij}|i\rangle\langle j|$$

Where $|i\rangle\langle j|$ is the matrix that has a 1 at the position (i, j) and 0 everywhere else.

Tensor product of matrices

Given two Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ we know how to compute the inner product space of the tensor product between our two spaces

$\mathcal{H}_1, \mathcal{H}_2$ as seen before. What we want to see now is the tensor product between two matrices A and B.

To do so we'll use the definition of the matrix using the '*ket bra*' notation of the definition above.

$$A \otimes B = \sum_{i,j,j,l} A_{ij} B_{jl} \underbrace{|i\rangle\langle j| \otimes |k\rangle\langle l|}_{(|i\rangle \otimes |k\rangle)(\langle j| \otimes \langle l|)}$$

Remark 1. The notation $|i\rangle \otimes |k\rangle$ can be simplified by $|i, k\rangle$.

$$A \otimes B = \sum_{i,j=1}^{D_1} \sum_{k,l}^{D_2} A_{ijkl} |i, k\rangle\langle j, l|$$

The resulting dimension of our matrix is given by $D_1 D_2 \times D_1 D_2$ where

$$(A \otimes B)_{ik,jl} = A_{ij} B_{kl}$$

Example

Given the product of $\mathbb{C}^2 \otimes \mathbb{C}^2$, let us see what would be the tensor product of the two matrices:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

The goal is to compute $A \otimes B$.

$$\begin{aligned} A \otimes B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ &= \begin{pmatrix} a \cdot B & bB \\ cB & dB \end{pmatrix} \\ &= \begin{pmatrix} ae & af & be & bf \\ ad & ah & bd & bh \\ ce & cf & ge & gf \\ cd & ch & gd & gh \end{pmatrix} \end{aligned}$$

2.0.2 Hermitian matrix

First let us do some notation:

$$\begin{aligned} A^\dagger &= \sum_{i,j} A_{ij}^* |j\rangle\langle i| \\ &= \sum_{i,j} A_{ij}^* |i\rangle\langle j| \end{aligned}$$

We have the following:

$$(A^*)_{ij} = A_{ij}^*$$

Hermitiam matrices

Definition 6. An Hermitian matrix is a given operator A on an Hilbert space $\mathcal{H}$ where

$$A = A^\dagger$$

Unitary matrices

Definition 7. An unitary matrix is an Hermitian operator U on an Hilbert space $\mathcal{H}$ where

$$U^{-1} = U^{\dagger}$$

Here an equivalent property can be derived

$$UU^{\dagger} = U^{\dagger}U = I$$

Which is equivalent to say for two any given pure states $|\varphi\rangle, |\varphi\rangle$

$$\langle\varphi||\varphi\rangle = \langle\varphi|UU^{\dagger}|\varphi\rangle = (U|\varphi\rangle)^{\dagger} \cdot (U|\varphi\rangle)$$

Spectral theorem for Hermitian matrices

Theorem 1. Given a Hermitian matrices A we have that all eigenvalues of our matrix are real number where their respective eigenvector are orthonormal.

$$A|u_i\rangle = \lambda_i|u_i\rangle$$

with $\lambda_i \in \mathbb{R}$.

In particular, we have:

$$A = \sum_{i=1}^{d} \lambda_i |v_i\rangle\langle v_i|$$

2.1 Principles of Quantum Physics

In this section, we will introduce the principles in an axiomatic form. We will formulate them in algebraic manner appropriate for quantum information and computation in finite dimensional Hilbert space.

Hilbert space

Note here that we have not defined what a Hilbert space here is so let me define it (wikipedia)

Definition 8. (Hilbert Space) A **Hilbert** space is a real or complex inner product space that is also a complete metric space with respect to the distance function induced by the inner product

To say that a complex vector space $\mathcal{H}$ is a complex inner product space means that there is an inner product $\langle x, y\rangle$ associating a complex number to each pair of elements x, y of $\mathcal{H}$ that satisfies the following properties:

1. The inner product is conjugate symmetric; that is the inner product of a pair of elements is equal to the complex conjugate of the inner product of the swapped elements:

   $$\langle y, x\rangle = \overline{\langle x, y\rangle}$$

   Importantly, this implies that $, x\rangle$ is a real number
2. The inner product is linear in its first argument. For all complex number $a, b \in \mathbb{C}$

   $$\langle ax_1 + bx_2, y\rangle = a\langle x_1, y\rangle + b\langle x_2, y\rangle$$

3. The inner product of an element with itself is positive definite

   $$\begin{aligned} \langle x, x\rangle &> 0 \qquad \text{if } x \neq 0 \\ \langle x, x\rangle &= 0 \qquad \text{if } x = 0 \end{aligned}$$

2.1.1 Principle 1: State vectors

The state of a physical system can be described (isolated from the rest of the universe) as a vector in a finite dimensional Hilbert space $\mathcal{H}$, we denote it as

$$|\psi\rangle \in \mathcal{H}$$

This vector $|\psi\rangle$ must be normalized with respect to the inner product of that Hilbert space, meaning $\langle\psi|\psi\rangle = 1$.

Remark 2. The principle doesn't say how to choose $|\psi\rangle$ neither the space itself $\mathcal{H}$. This decision comes from the underlying physics of the system.

Remark 3. In a non-isolated system we will have to generalize this principle. This will be done in later part of the course when introducing the notion of **density** matrix.

Hence let us start with the most basic example in quantum physics, the qbit! The Hilbert space describing a qbit is given as $\mathcal{H} = \mathbb{C}^2$ which can be denoted as

$$\begin{aligned}\mathcal{H} = \mathbb{C}^2 &= \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \alpha, \beta \in \mathbb{C} \right\} \\ &= \{\alpha|0\rangle + \beta|1\rangle, \alpha \in \mathbb{C}, \beta \in \mathbb{C}\}\end{aligned}$$

Furthermore for this state to be correctly the state of a qbit it also must satisfy normalization, meaning its norm is 1. Recall not the definition of the inner product of a Hilbert space did we even define our inner product here? Recall that we need to validate the three properties above, a valid choice here is the following inner product:

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i^* y_i$$

You can check that this inner product here fulfils all properties. Going back on our requirement ($\|x\| = 1$) we will therefore need

$$\alpha^*\alpha + \beta^*\beta = 1$$

With that requirement, we have the Hilbert space of a qbit.

Also in terms of notation, we will often use the letter ψ, φ and φ in order to represent quantum states. We will also use the term state to refer to quantum state as all work here is primarily done in the quantum world. Hence we define our qbit as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

For more general states, ones of dimension d (i.e. $\mathcal{H} = \mathbb{C}^d$) our state is constructed as

$$|\psi\rangle = \sum_{i=1}^{d} \alpha_i|i\rangle$$

Remark 4. Note that state doesn't have to be necessarily finite. A good example of this are quantum harmonic oscillators (so PHYS-344) in which our state is defined as

$$|\Psi\rangle = \sum_{n=0}^{\infty} \psi_n|n\rangle$$

2.1.2 Principle 2: Unitary evolution on time evolution

The state of an isolated system evolves (as a function of time) unitarily. This means that of $|\psi_0\rangle$ is a state at time $= 0$, the state $|\psi_t\rangle$ is given by

$$U_t|\psi_0\rangle = |\psi_t\rangle$$

Where $U_t^\dagger U_t = U_t U_t^\dagger = I$ i.e. U_t is unitary (usually time dependent) matrix

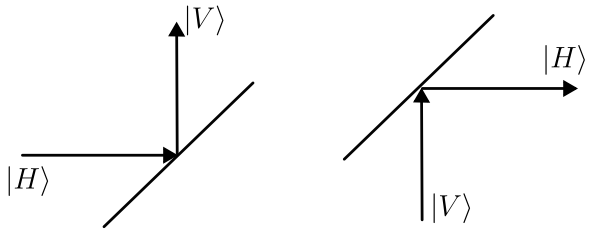
Remark 5. One can show that unitary matrix represent rotation of the space by a specified angles.

Remark 6. For people of have done or who are doing MATH-310 (Algebra), one can see that the time evolution matrices form a group in the sense that

$$U_{t_2}U_{t_1}U_{t_1+t_2} \qquad , U_{t=0} = I$$

Example

Let us take the example of a reflecting mirror :



The physical process which transforms the incident ray into the reflected ray is described by the unitary matrix:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |H\rangle\langle V| + |V\rangle\langle H|$$

For $|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Let us take a step back on what is actually going on here. What a perfectly reflecting mirror with a tilted angle here reflect a ray that is normal ($\langle\psi_{t_1}|\psi_{t_2}\rangle = 0$) for the state after and before the mirror, so here we would require that $U|H\rangle = |V\rangle$ and vis versa which if you apply X on any of those two states gives the intended results.

2.2 Analysis of MZ interferometer 1

Let us recall the experimental setup, first the Hilbert space: $\mathcal{H} = \mathbb{C}^2 = \{\alpha|h\rangle + \beta|v\rangle, \alpha, \beta \in \mathbb{C}\}$. Where $|h\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |v\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Here the state represent the '*path state*' of the given photon. Here the path state must also satisfy the normalization of the vector

$$|\alpha|^2 + |\beta|^2 = 1$$

The input state here is simply $|h\rangle$. The goal here is to see how this input state evolve in time. To do so let us recall the second principle which state that the evolution is given by a unitary matrices (when not measured). Hence the first semi-transparent mirror is represented by the **Hadamard** matrix:

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|h\rangle\langle h| + |h\rangle\langle v| + |v\rangle\langle h| - |v\rangle\langle v|)$$

And as we can see we effectively have $HH^\dagger = H^\dagger H = I$.

The output of the semi-transparent mirror is given by

$$H|h\rangle = \frac{1}{\sqrt{2}}(|h\rangle + |v\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

Perfect mirror

Let us continue with the second principle and see what the perfect mirrors do to our current quantum state. To see what they do let us first define them

$$X = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix} = |h\rangle\langle v| + |v\rangle\langle h|$$

As we can see, this unitary matrix behave as a logical negation meaning taking $X|h\rangle$ gives $|v\rangle$ and vis versa.

Output of the reflection

Now let us see what is the output of the two mirrors together

$$XH|h\rangle = \frac{1}{\sqrt{2}}(X|h\rangle + X|v\rangle) = \frac{1}{\sqrt{2}}(|v\rangle + |h\rangle)$$

Finally, let us see what is the final state after the final semi-transparent mirror. As the first mirror, this one is also represented using the same unitary matrix H. To see the final state, we as before are required to apply H to the current state which gives the following

$$HXH|h\rangle = \frac{1}{\sqrt{2}}(H|v\rangle + H|h\rangle) = \frac{1}{\sqrt{2}}\left(\frac{|h\rangle - |v\rangle}{\sqrt{2}}\,\frac{|h\rangle + |v\rangle}{\sqrt{2}}\right) = |h\rangle$$

Remark

The wave interpretation here would explain this as making the D_2 output being a destructive wave addition wheras the output into D_1 is constructive.

Principle 4

Recall the principle: *Measurement is done in a basis*, we will use $\{|h\rangle, |v\rangle\}$. Using this basis, one can use Born rule to compute the probability of each one being selected as

$$P(|h\rangle \text{ being selected}) = P(D_1, \text{clicks}) = |\langle h|h\rangle|^2 = 1$$
$$P(|v\rangle \text{ being selected}) = P(D_2, \text{clicks}) = |\langle v|h\rangle|^2 = 0$$

Remark 7. Let us first recall principle 3 which states that an **Observable** or quantity that I measure can be represented using a hermitian operator where each eigenvalues are an output and its correspondent eigenvector be the vector to compute the probability of the given output.

In our case the observable which can be called C is defined the following way

$$C = (+1)|h\rangle\langle h| + (-1)|v\rangle\langle v| = \begin{pmatrix}+1 & 0\\0 & -1\end{pmatrix}$$

2.3 MZ interferometer 2

Here we now add **dephaser** which also are unitary matrices namely

$$P_1 = \begin{pmatrix}e^{i\varphi_1} & 0\\0 & 1\end{pmatrix}$$
$$P_2 = \begin{pmatrix}1 & 0\\0 & e^{i\varphi_2}\end{pmatrix}$$

Using those dephaser,the mirror now are also changed, namely

$$U_{st} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$
$$U_{Riff} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

As we can check, both P_i matrices here share the same eigenbasis which implies that they are commutative which mean that the evolution in time can be computed the following way

$$U_{st}U_{Riff}P_1P_2U_{st}|h\rangle = -\frac{e^{i\varphi_2} + e^{i\varphi_2}}{2}|h\rangle + i\left(\frac{e^{i\varphi_1} - e^{i\varphi_2}}{2}\right)|v\rangle$$

Hence the probabilities are computed as

$$P(D_1 \text{ clicks}) = \left|\frac{e^{i\varphi_2} + e^{i\varphi_2}}{2}\right|^2 = \cos^2\left(\frac{\varphi_1 - \varphi_2}{2}\right)$$
$$P(D_2 \text{ clicks}) = \left|\frac{e^{i\varphi_2} - e^{i\varphi_2}}{2}\right|^2 = \sin^2\left(\frac{\varphi_1 - \varphi_2}{2}\right)$$

In this case, we can also define the following observable matrix

$$C' = (+1)|+\rangle\langle+| + (-1)|-\rangle\langle-|$$

2.4 From electromagnetic to quantum mechanics

The way we have defined our qbit only took into account two directions. However why should we limit ourself to this? Polarization degree of freedom of a photons, spin of a particles can also be perfectly represented as qbits.

Electromagnetic field

First let us recall what is an electromagnetic field is, using Maxwell equations, the field can be defined using a magentic and electric field $\vec{B}$ and $\vec{E}$ respectively which vibrate into a plane normal to a given $\vec{z}$ direction. Using Maxwell equation, one can deduce $\vec{B}$ using only $\vec{E}$. Hence the field can be defined only using $\vec{E}$ which cen be defined the following way

$$\vec{E} = \text{Re}\left\{\vec{E}_0 e^{i(kz-\omega t)}\right\}$$

Where the $\vec{E}_0$ set the direction of the vibration of the field and the exponent defines the polarization. Which when taking this into a vector notation, we obtain

$$E_0\begin{pmatrix} \cos\theta \\ \sin\theta e^{i\varphi} \\ 0 \end{pmatrix}$$

Photons also retain a sort of '*pole state*' defined as $\begin{pmatrix} \cos\theta \\ \sin\theta e^{i\varphi} \end{pmatrix} \in \mathbb{C}^2 = \mathcal{H}$. As we can see here, we would like to define our state using the x-y plane inside a plane. However this $e^{i\varphi}$ gives us a problem here, what this factor implies physically is that the photon also disturb the $z-$plane.

To translate this into Dirac notation, we define it the following way

$$\cos\theta|0\rangle + \sin\theta e^{i\varphi}|1\rangle$$

Example

If for instance we define the case where $\varphi = \frac{\pi}{2}$ and $\theta = \frac{\pi}{4}$ we get the following state

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

Which as we can see implies that the polarization given by the $e^{i\varphi}$ term can either rotate in both direction. Here as $\varphi \neq 0$, we have that the blabla is rotating around the z axis. The φ given here represent the polarization degree of freedom of our photon.

2.5 Concept of Entanglement

Let us define our total Hilbert space as $\mathcal{H}_{\text{total}} = \mathcal{H}_a \otimes \mathcal{H}_b$ (e.g. two quantum bits together $\mathcal{H}_a = \mathbb{C}^2, \mathcal{H}_b = \mathbb{C}^2$).

Definition 9. Product state $|\varphi\rangle$ are such that

$$\exists |\varphi_a\rangle \in \mathcal{H}_a, |\varphi_b\rangle \in \mathcal{H}_b$$

such that $\mathcal{H}_{\text{total}} \in |\varphi\rangle = |\varphi_a\rangle \otimes |\varphi_b\rangle$

What we mean is that a state is a product state if our state $|\varphi\rangle$ can be written as a tensor product between to state from each Hilbert space $\mathcal{H}_a$ and $\mathcal{H}_b$.

Definition 10. **Entangled states** are states which cannot be written as a tensor product of two state, state that are not product state.

Example

If we take for instance the following state inside $\mathbb{C}^2 \otimes \mathbb{C}^2$:

$$\begin{aligned} |\varphi\rangle = |+\rangle \otimes |-\rangle &= \frac{|0\rangle + |1`\rangle}{\sqrt{2}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) \end{aligned}$$

Here we have that $|\varphi\rangle$ is a product state by construction.

Let us for instance change a minus sign in our state giving

$$|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$$

To prove that the state $|\psi\rangle$ is not a product state, we can do the following by contradiction. Assume that our state is entangled, then we know that there exists four coefficients such that

$$|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$$

Here one can just develop our four equalities giving us the following system

$$\begin{cases} \alpha\gamma = \frac{1}{2} \\ \alpha\delta = -\frac{1}{2} \\ \beta\gamma = \frac{1}{2} \\ \beta\delta = \frac{1}{2} \end{cases}$$

If we were working with real number, this can be proven fairly easily as the product of the LHS would be positive whereas the RHS would me negative ($-\frac{1}{16}$). But as we are working with complex number one should show it differently, what we need to do here is to make the modulus pop out. To do so, one can just take the conjugate as we have that $\overline{\alpha\gamma} = \frac{1}{2}$ and that $\overline{\beta\delta} = \frac{1}{2}$. Which when now multiplying everything together gives

$$|\alpha|^2|\beta|^2|\gamma|^2|\delta|^2 = -\frac{1}{16}$$

Which is a contradiction.

2.5.1 Bell states

There are four states that plays a special roles in quantum communication and information theory, namely in teleportation, dense coding but also in QKD and many more topics. Those state are the '*maximum entangled*'

What are those? The bell states form an orthogonal basis of the state of two qbits $\mathbb{C}^2 \otimes \mathbb{C}^2$. Those state are given as

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B)$$
$$|B_{01}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B + |1\rangle_A \otimes |0\rangle_B)$$
$$|B_{10}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B - |1\rangle_A \otimes |1\rangle_B)$$
$$|B_{11}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |1\rangle_B - |1\rangle_A \otimes |0\rangle_B)$$

Here, one can show here that those states are entangled.

Computational orthogonal basis Let us also define the computational orthogonal basis of our new space $\mathbb{C}^2 \otimes \mathbb{C}^2$ namely

$$|0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \qquad |0\rangle \otimes |1\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$$
$$|1\rangle \otimes |0\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} \qquad |1\rangle \otimes |1\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

Important property of $|B_{00}\rangle$ Let us define the state $|\theta\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$ which is the polarization state of a photon by θ. Then we can also define the perpendicular state

$$|\theta_\perp\rangle = -\sin\theta|0\rangle + \cos\theta|1\rangle$$

Theorem 2.

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|\theta\rangle_A \otimes |\theta\rangle_B + |\theta_\perp\rangle \otimes |\theta_\perp\rangle_B)$$

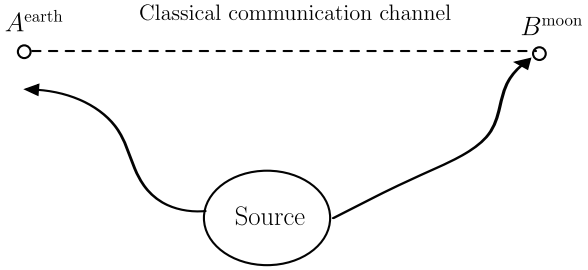
What this implies is that the entanglement here $|B_{00}\rangle$ is angle independent. In Alice's lab, the angle of our state locally is sort of '*random*' but is the **same** as the state inside Bob's lab. What we mean here by that which we will see later is that the entropy of our angle in both labs are maximized.

Remark The prove for all θ is left to the reader.

2.5.2 Teleportation

Assume that we have Alice A^{earth} on earth which has a photon in a state $|\varphi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$ where $\alpha, \beta \in \mathbb{C}$ are unknown.

Protocol



Given our initial state $|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}$ the goal of the protocol is to '*teleport*' the state $|\varphi\rangle$ from A to B, $|\varphi\rangle_1|\varphi\rangle_3$ where we have the entangled state $|B_{00}\rangle$ being split with one qbit to Alice $|\varphi\rangle_2$ and the other to Bob $|\varphi\rangle_3$ (note that we say separate in the sense that one qbit is at Alice's location and the other at Bob's location and not as two separate states).

First step Alice does a local measurement of her two qbits with apparatus that's equivalent to a Bell basis.

Second step Alice observes her result, one among $|B_{00}\rangle, |B_{01}\rangle, |B_{10}\rangle, |B_{11}\rangle$ and sends two classical bits of information through the classical channel.

Third step Bob receives the classical message $ij \in \{00, 01, 10, 11\}$ and will do a local unitary operation on his qbit.

- If he receives $00 \to I_3$
- $01 \to X_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- $10 \to Z_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- $11 \to Z_3X_3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Then Bob receives exactly $|\varphi\rangle_3 = \alpha|0\rangle_3 + \beta|1\rangle_3$

Analysis

The measurement projects the state of Alice, suppose for the analysis that she get $|B_{01}\rangle$ (the same argument can be done for any other bell states), what this implies is that our state is projected onto $|B_{01}\rangle$ using the following projector

$$|B_{01}\rangle_{12}\langle B_{01}|_{12}$$

In Bob's lab nothing has happened which implies that the first operation is a construction using the above projector and the identity applied to our state which gives the following

$$(|B_{01}\rangle_{12}\langle B_{01}|_{12} \otimes I_3)(|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}) = |B_{01}\rangle_{12} \otimes \underbrace{\langle B_{01}|_{12}|\varphi\rangle_1 \otimes |B_{00}\rangle_{23}}_{\text{state vector for 3 in Bob's lab}}$$

As we can see, there is not entanglement now between 2 and 3 but 1 and 2 are now entangled. Let us now compute the right par of the RHS which gives us

$$\begin{aligned}
&\frac{1}{\sqrt{2}}(\langle 0|_1 \otimes \langle 1|_2 + \langle 1|_1 \otimes \langle 0|_1)(\alpha|0\rangle_1 + \beta|1\rangle_1) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 \otimes |0\rangle_3 + |1\rangle_2 \otimes |1\rangle_3) \\
&= \frac{1}{\sqrt{2}}(\alpha\langle 1|_2 + \beta\langle 0|_2) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 \otimes |0\rangle_3 + |1\rangle_2 \otimes |1\rangle_3) \\
&= \frac{1}{2}(\alpha|1\rangle_3 + \beta|0\rangle_3)
\end{aligned}$$

But as we didn't normalized after the measurement we should do it now giving us

$$= \alpha|1\rangle_3 + \beta|0\rangle_3$$

Then after this he receives from Alice the two bits 01 which means that he needs to apply X_3 giving us

$$\begin{aligned} X(\alpha|1\rangle_3 + \beta|0\rangle_3) &= \alpha X|1\rangle + \beta X|0\rangle \\ &= \alpha|0\rangle + \beta|1\rangle = |\varphi\rangle_3 \end{aligned}$$

2.5.3 Dense coding

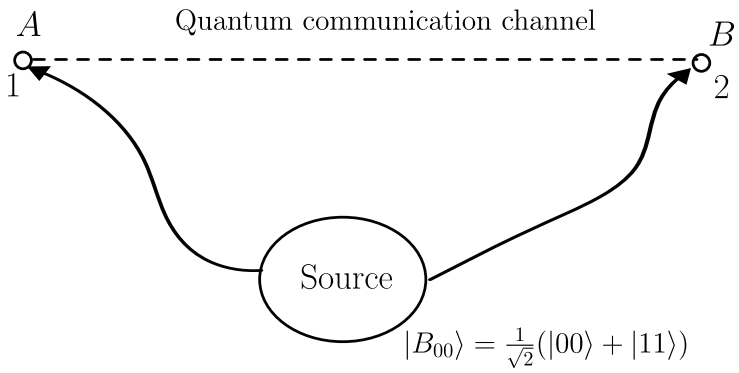


Figure 2.1: Dense coding protocol

The goal is that Alice transmits two classical bits to Bob using only one qbit.

Protocol

If Alice wants to send 00, A will sends with I_1 and send the qbit to Bob $\Longrightarrow$ Bob receives full EPR pair $|B_{00}\rangle_{\text{Bob}}$

- If Alice want to send 01, Alice will apply X_1 and send the qbit to Bob $\Longrightarrow$ Bob receives $(X_1 \otimes I_2)|B_{00} = |B_{01}\rangle\rangle_{\text{Bob}}$
- If Alice want to send 10, Alice will apply Z_1 and send the qbit to Bob $\Longrightarrow$ Bob receives $(Z_1 \otimes I_2)|B_{00} = |B_{10}\rangle\rangle_{\text{Bob}}$
- If Alice want to send 11, Alice will apply Z_1X_1 and send the qbit to Bob $\Longrightarrow$ Bob receives $(Z_1X_1 \otimes I_2)|B_{00} = |B_{11}\rangle\rangle_{\text{Bob}}$

Finally, Bob Measure the full pair in the Bell basis which the output will be $|B_{ij}\rangle_{\text{received}}$ Where ij is the bits that Alice sent.

2.6 Bell Inequalities

In this section we will cover only one inequality, the CHSH inequality which was first proposed in 1969. Let us first start with experimental setting,
Alice does measurements with the following basis :

$$\begin{cases} |\alpha\rangle = \cos\alpha|0\rangle + \sin\alpha|1\rangle \\ |\alpha_\perp\rangle = -\sin\alpha|0\rangle + \cos\alpha|1\rangle \end{cases}$$

$$\left\{|\alpha'\rangle, |\alpha'_\perp\rangle\right.$$

The outcome of those measurements is given as random variables

$$a = \pm 1$$
$$a' = \pm 1$$

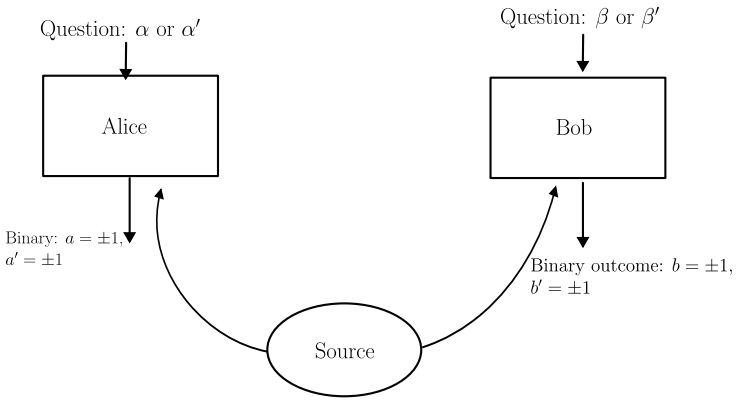


Figure 2.2: Experimental Setting

Bob also measures with a basis

$$\{|\beta\rangle, |\beta_\perp\rangle\}$$
$$\{|\beta'\rangle, |\beta'_\perp\rangle\}$$

With the outcome of the measurements being also random variables

$$b = \pm 1$$
$$b' = \pm 1$$

Which gives four possibilities

$$(\alpha, \beta) \to (a, b)$$
$$(\alpha', \beta) \to (a', b)$$
$$(\alpha, \beta') \to (a, b')$$
$$(\alpha', \beta') \to (a', b')$$

Then after this they collect their strings A and B and communicate and exchange their results. After having collected everything, they will compute the **correlation coefficient** which is given as (knowing $N_1 \approx N_2 \approx N_3 \approx N_4 \approx \frac{N}{4}$)

$$X^{\text{experimental}} = \frac{1}{N_1} \sum_{i_1:(\alpha,\beta)} a_{i_1} b_{i_1} + \frac{1}{N_2} \sum_{i_2:(\alpha,\beta')} a_{i_2} b'_{i_2} - \frac{1}{N_3} \sum_{i_3:(\alpha',\beta)} a'_{i_3} b_{i_3} + \frac{1}{N_4} \sum_{i_4:(\alpha',\beta')} a'_{i_4} b'_{i_4}$$

Theoretical prediction

The theoretical prediction of our experiment is given as

$$X^{\text{theory}} = \mathbb{E}_{\alpha\beta}(ab) + \mathbb{E}_{\alpha\beta'}(ab') - \mathbb{E}_{\alpha'\beta}(a'b) + \mathbb{E}_{\alpha'\beta'}(a'b')$$

To be able to compute this we need to introduce the hidden variable local theory in order to give a view point of a '*classical physicist*'.

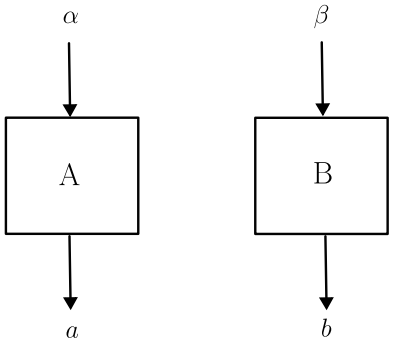
What we are saying is that the outcome of a measurement of α into a is given by the probability

$$P_A(a|\alpha, \lambda)$$

Where λ is a collection of hidden variables, meaning that the outcome distribution is determined by λ and the setting α. What's important here is also the '*local*' word, the result of a does not depend on β, the experiment of Alice is not influenced by any other factor.

The same mechanism works with Bob giving

$$P_B(b|\beta, \lambda)$$



Which can be formalized as

$$P(a,b|\alpha,\beta,\lambda) = P_A(a|\alpha,\lambda)P_B(b|\beta,\lambda)$$

We also assume that the hidden variables are distributed according to some distribution $h(\lambda)$ which does **not** depend on the settings α, β (**measurement independence** or '*free choice*'). This is the third standard assumption alongside locality and realism (hidden variables): without it, we could not use the same $h(\lambda)$ in all four expectations.

Theorem 3. CHSH inequality (Bell inequality)

$$-2 \leq X^{\text{LHV theory}} \leq 2$$

Proof. *Let us first try to compute $\mathbb{E}_{\alpha,\beta}(ab)$. To do so we first need to compute the conditional over λ times the probability of λ which implies integrating over λ:*

$$\begin{aligned}\mathbb{E}_{\alpha,\beta}(ab) &= \sum_{a,b=\pm 1} ab \int d\lambda P(a,b|\alpha,\beta,\lambda)h(\lambda) \\ &= \sum_{a,b=\pm 1} ab \int d\lambda P_A(a|\alpha,\lambda)P_B(b|\beta,\lambda)h(\lambda)\end{aligned}$$

Note that $h(\lambda) \geq 0$ and $\int h(\lambda)d\lambda = 1$, also adding the probability distribution of a' and b' (which shouldn't change anything as their sums are 1).

$$= \sum_{a,a',b,b' \in \{\pm 1\}} ab \int d\lambda P_A(a'|\alpha',\lambda)P_A(a|\alpha,\lambda)P_B(b'|\beta',\lambda)P_B(b|\beta,\lambda)h(\lambda)$$

Let us now define

$$Q(a,a',b,b'|\alpha,\alpha',\beta,\beta') = \int d\lambda P_A(a|\alpha,\lambda)P_A(a'|\alpha',\lambda)P_B(b|\beta,\lambda)P_B(b'|\beta',\lambda)h(\lambda)$$

Let us simplify the notation by removing the conditional for convenience (they are still there), which implies that our expected value $\mathbb{E}_{\alpha\beta}(ab)$ can be rewritten as

$$\mathbb{E}_{\alpha,\beta}(ab) = \sum_{a,a',b,b'} ab\, Q(a,a',b,b')$$

The key idea is that Q is a genuine joint probability distribution over all four outcomes: $Q \geq 0$ and $\sum_{a,a',b,b'} Q(a,a',b,b') = 1$. Locality plus hidden variables guarantees that such a joint distribution exists, even though a and a' (or b and b') can never be measured together. This is the whole content of the theorem. Which means that our random variable can be rewritten as

$$\begin{aligned}X^{LHV\ theory} &= \sum_{a,a',b,b'} Q(a,a',b,b')[ab + ab' - a'b + a'b'] \\ &= \sum_{a,a',b,b'} Q(a,a',b,b')[a(b+b') + a'(b'-b)]\end{aligned}$$

Notice here that all random variables here are binary which means that $(b+b')$ *can either be* $0, 2$ *or* -2*. Notice now that the case* $b = b'$ *gives* $2ab + 0 = \pm 2$*, and if* $b \neq b'$ *(e.g.* $b = 1, b' = -1$*) we get* $0 - 2a' = \pm 2$*. We can convince ourselves that the result of the* $[\cdot] = \pm 2$*. Therefore*

$$\left|X^{LHV\ theory}\right| \leq \sum_{a,a',b,b'} Q(a,a',b,b')|[\cdot]| = 2 \sum_{a,a',b,b'} Q(a,a',b,b') = 2$$

□

We can show that the hidden variable local theory cannot be applied to quantum mechanics! Meaning that the joint distribution of our two states here cannot be factorized into a product of two probability distributions. Intuitively, this is related to Bell states or more generally entangled states (which cannot be factorized into tensor products). Concretely, we have that for EPR Bell pairs distributed by some

$$\lambda \leftrightarrow |B_{00}\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle)$$

We find that

$$\max_{\alpha,\beta,\alpha',\beta'} X^{\text{QM theory}}(\alpha,\beta,\alpha',\beta') = 2\sqrt{2}$$

Which therefore implies that quantum mechanics cannot be represented using the hidden variable local theory, meaning technically that quantum mechanics is **non-local**.

Quantum prediction for Bell pairs (EPR pairs)

Let us now look at the same '*experiment*' but with Bell pairs, we have a source of qubits for $i = 1, \ldots, N$ with a distribution $|B_{00}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B)$ with A and B observables defined as

$$A = (+1)|\alpha\rangle\langle\alpha| + (-1)|\alpha_\perp\rangle\langle\alpha_\perp|$$
$$A' = (+1)|\alpha'\rangle\langle\alpha'| + (-1)|\alpha'_\perp\rangle\langle\alpha'_\perp|$$

And the same for Bob

$$B = (+1)|\beta\rangle\langle\beta| + (-1)|\beta_\perp\rangle\langle\beta_\perp|$$
$$B' = (+1)|\beta'\rangle\langle\beta'| + (-1)|\beta'_\perp\rangle\langle\beta'_\perp|$$

Our goal is to compute the expected value as in the classical case which is given as

$$\begin{aligned} X^{\text{QM theory}} &= \langle B_{00}|A \otimes B|B_{00}\rangle + \langle B_{00}|A \otimes B'|B_{00}\rangle - \langle B_{00}|A' \otimes B|B_{00}\rangle + \langle B_{00}|A' \otimes B'|B_{00}\rangle \\ &= \cos(2(\alpha-\beta)) + \cos(2(\alpha-\beta')) - \cos(2(\alpha'-\beta)) + \cos(2(\alpha'-\beta')) \end{aligned}$$

The proof is left as an exercise. Then what is left to do is to see if our $X^{\text{QM theory}}$ is contained in the interval that we have told before ($\left[-2\sqrt{2}, 2\sqrt{2}\right]$). If we maximize our function with respect to $\alpha, \alpha', \beta, \beta'$ what we get is

$$\alpha = 0 \qquad \alpha' = -\frac{\pi}{4}$$
$$\beta = \frac{\pi}{8} \qquad \beta' = -\frac{\pi}{8}$$

Which gives

$$\max X^{\text{QM theory}} = 2\sqrt{2}$$

Okay so we see that formalizing as before doesn't work as well, a question that is arising is the following :

$$P(a,b|\alpha,\beta,\text{given Bell states}) =?$$

Answer

$$= \frac{1}{4}(1 + ab\cos(2(\alpha-\beta))) \neq f(a|\alpha)f(b|\beta)$$

Which implies that the following distribution **cannot** be factorized. We therefore say that quantum mechanics is **non-local**.

Suppose that the source distributes a state $|\varphi_A\rangle \otimes |\varphi_B\rangle$, we ask ourselves what should be $X^{\text{QM, tensor prod}}$? Can this be bounded by ± 2? (exercise)

2.7 Ekert (1991) QKD protocol

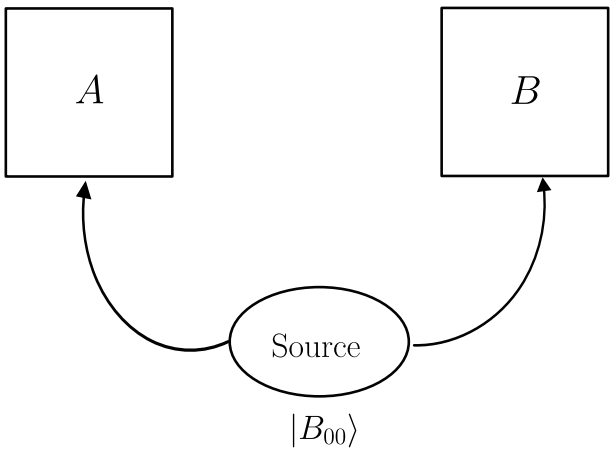


Figure 2.3: Ekert protocol

- Alice and Bob keep secret the basis in which they will do the measurement and have either 0 or 1 giving $x_i = 0, 1$ and $y_i = 0, 1$.

Alice will have the angles

$$\alpha_1 = -\frac{\pi}{4}$$
$$\alpha_2 = -\frac{\pi}{8}$$
$$\alpha_3 = 0$$

Bob will have

$$\beta_1 = -\frac{\pi}{8}$$
$$\beta_2 = 0$$
$$\beta_3 = -\frac{\pi}{8}$$

Alice and Bob reveal the basis choices for the angles $\alpha_1, \alpha_3, \beta_1, \beta_3$. They compute $X^{\text{experimental}} = 2\sqrt{2}$ which is correct only if there is no eavesdropper. We can then produce an OTP by taking bits for angles

$$(\alpha_2, \beta_1) = \left(-\frac{\pi}{8}, -\frac{\pi}{8}\right)$$
$$\text{and } (\alpha_3, \beta_2) = (0, 0)$$

In which we have $x_i = y_i$.

2.8 References

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